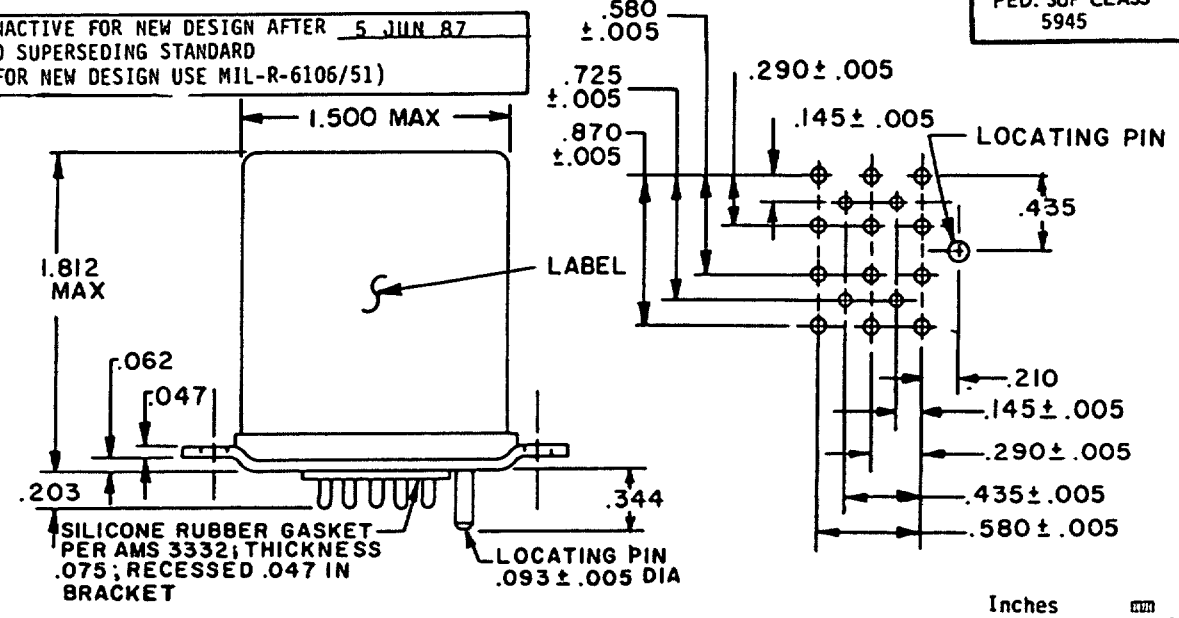
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User activities: Army - Navy - Air Force -

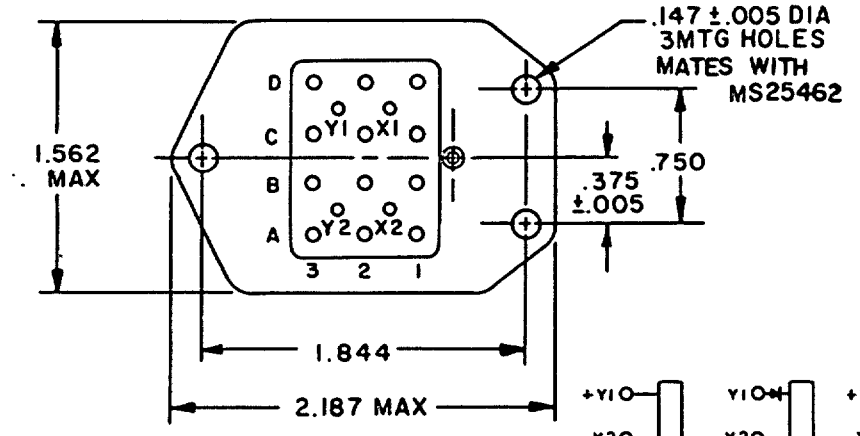
Review activities: Army - Navy - EC Air Force - 11

INACTIVE FOR NEW DESIGN AFTER 5 JUN 87
NO SUPERSEDING STANDARD
(FOR NEW DESIGN USE MIL-R-6106/51)

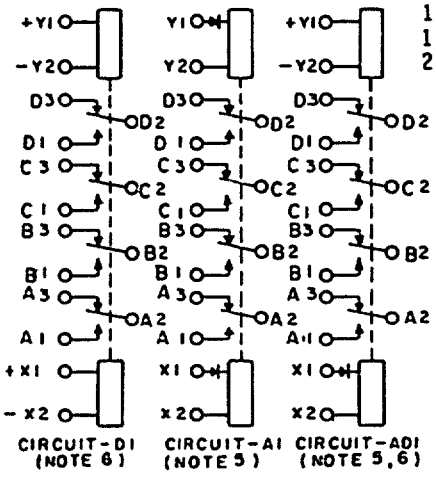
FED. SUP CLASS 5945



12 PINS, .0625 ± .0010 DIA
4 PINS .040 ± .001 DIA (X1, X2, Y1, AND Y2)
GOLD PLATED, MIL-G-45204, TYPE II, CLASS 1, UNDERPLATING, NICKEL 50 TO 150 MICROINCHES THICK



Inches	mm
.000	0.00
.001	0.03
.005	0.13
.040	1.02
.047	1.19
.062	1.57
.0625	1.59
.093	2.36
.145	3.86
.147	3.73
.203	5.16
.290	7.37
.344	8.74
.375	9.53
.435	11.05
.580	14.73
.725	18.42
.870	22.10
1.500	38.10
1.562	39.67
1.812	46.02
1.844	46.84
2.187	55.55



APPROVED 1 Nov 1960 REVISED (F) 6 JUN 87

(F) ENTIRE STANDARD REVISED

P.A	International Interest	TITLE	MILITARY STANDARD
AIR FORCE - 85 Other Cust Navy - AS		RELAYS, ELECTROMAGNETIC, 10 AMPERES, 4 PDT, TYPE I, MAGNETIC LATCH, SOCKET MOUNTED, HERMETICALLY SEALED	MS25461
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 1 OF 5

DD FORM 1 MAY 73 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. 5945-0745-18

AMSC N/A

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Terminal numbers need not appear on relay headers provided there is affixed to the relay a suitable legible circuit diagram that permanently and positively identifies each terminal location specified herein.
5. The use of diodes on ac relays is optional. Actual application must be shown on label.
6. Relay is magnetically latched in both positions. Caution note to observe polarity must appear on relays with dc coils.
7. In the event of confilict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
8. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.
9. Shock, vibration, and acceleration requirements application with coils de-energized.
10. Pins to be perpendicular to header surface within one degree

TABLE I. Dash numbers and characteristics.

Dash number MS 25461-	Type	Coil	Terminal type	Max weight in pounds
D1	I	dc	Plug in	0.40
A1	I	ac	Plug in	0.42
AD1	I	ac-dc	Plug in	0.42

APPROVED 1 Nov 1960
REVISED (F) ENTIRE STANDARD REVISED

User activities: Army - Navy - Air Force -
Review activities: Army - Navy - EC Air Force - 11
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Review activities: Army - Navy - EC Air Force - 11

User activities: Army - Navy - Air Force -

TABLE II. Operating characteristics.

MS part no. MS25461-	Coil data											Time - (milliseconds maximum)				
	Coil	Nominal			Max		Max pick-up voltage			Drop-out voltage	Operate 3/	Release 4/	Contact bounce			
		Volts 1/	Freq. Hz	Res Ω	Volts	Amperes	Normal 3/	High temp test	Cont current test				Main		Aux	
													NO	NC	NO	NC
D1	X1, X2 Y1, Y2	28	dc	N/A	29	0.17	18	18	19.8	N/A	25	N/A	2	2		
A1	X1, X2 Y1, Y2	115	400 5/	N/A	122	0.07	90	90	95	N/A	25	N/A	2	2		
AD1	X1, X2	115	400 5/	N/A	122	0.07	90	90	95	N/A	25	N/A	2	2		
	Y1, Y2	28	dc	N/A	29	0.17	18	18	19.8	N/A	25	N/A	2	2		

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
5/ MS25461-A1 and -AD1 may be used on 60 Hz if maximum ambient temperature is limited to +85°C (maximum coil current shall be 0.077 ampere).

DD FORM 1 MAY 73 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

Procurement Specification MIL-R-6106

AIR FORCE-85
Other Cust NAVY-AS

International interest

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 4 PDT, TYPE 1, MAGNETIC LATCH, SOCKET MOUNTED, HERMETICALLY SEALED

MILITARY STANDARD

MS25461

PAGE 3 OF 5

FED. SUP CLASS 5945

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User activities: Army - Navy - Air Force -

TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles X 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	10	10			10	6			10	6			
Inductive	100													
Inductive	20	6	6			10	4			10	4			
Motor	100	4	4			4	3			4	3			
Lamp	100	2	2			2	1.5			2	1.5			
Transfer load														2/
Mechanical life reduced current	400	2.5	2.5			2.5	2			2.5	2			
Intmd current	Applicable per specification													

1/ Absence of value indicates relay is not rated for 3-phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

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P.A AIR FORCE-85 Other Cust NAVY - AS

Procurement Specification MIL-R-6106

International interest

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 4 PDT, TYPE I, MAGNETIC LATCH, SOCKET MOUNTED, HERMETICALLY SEALED

MILITARY STANDARD MS 25461

PAGE 4 OF 5

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MILITARY STANDARD

MS25461

PAGE 5 OF 5

FED. SUP CLASS 5945

Environmental characteristics

Temperature range	-70°C to +125°C
Max altitude rating	80,000 ft
Shock G-level	50 G
Duration	11 ms
Max duration contact opening	10 µs
Vibration - sinusoidal	
G-level	10 G
Frequency range	5 - 1500 Hz
Acceleration	15 G

Electrical characteristics

Insulation resistance, initial	100 megohms
After life or environmental tests	50 megohms

Dielectric strength initial (sea level)

	Initial	After life tests
Coil to case	1,000 V rms	1,000 V rms
Aux contacts		
All other points	1,500 V rms	1,125 V rms

Dielectric strength (altitude)

	80,000 ft
Coil to case	250 V rms
Aux contacts	
All other points	350 V rms

Max contact drop initial	0.150 volt
After life test	0.175 volt
Overload current	40 amperes dc, 60 amperes ac
Rupture current	50 amperes dc, 80 amperes ac
Duty rating	Continuous
RFI specification	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

Quality conformance inspection

Performance of groups B and C tests are not applicable.

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